

Stereo Audio Distribution

In the 'Stereo Audio Distribution Buffer for Headphones' circuit published in July issue, the use of 100-ohm resistors (R11 through R18) in series with the output is incorrect. With a headphone of 32-ohm impedance, it will cause a 4x signal attenuation in the output (32/(100+32)). For minimum attenuation, its value should be 10 ohms or less.

Sanjay Jain
Through e-mail

The author Petre Tzo Petrov replies:

The resistor values of 100 ohms are correct. If you use around 100-ohm resistors, you can connect and disconnect one pair of headphones without having significant impact on the sound level of other headphones. The circuit has enough gain to compensate R11 through R18 up to around 330 ohms or even more. If you need a higher gain, you can increase the voltage gain of the circuit up to around 100 without significant problems with NE5532 or similar amplifier.

You can use 10-ohm value but:

1. The output amplitude will depend on the number of connected headphones or loads.

2. The output will be overloaded if you connect 10 ohms+32 ohms at all the four outputs, especially if you use a higher-voltage power supply, e.g., above 18V.

You can experiment with different loads and values for R11 through R18 and different signals sources and voltage gains to see the difference. But take care that they do not damage the output transistors when the values of R11 through R18 are too low! 100-ohm is on the safe side in most cases!

Electronics Projects Vol. 26

I am eagerly waiting for the release of Electronics Projects Vol. 26. Please update me. My subscription will soon expire. Please send some offers of kits for subscription to 'EFY Plus.'

Satish Uppal
Through e-mail

EFY: Electronics Projects Vol. 26 eBook will be available soon. For subscription and related offers, please see EFY ads in magazine.

From electronicsforu.com: Circuit Lab

The 'USB Charger for Mobile Phones' circuit published in December 2012 issue is a good idea!

Amresh

□ The 'Ding Dong Touch Bell' circuit published in February issue has impressed us!

Munish

□ The 'Cable Tester' circuit published in March issue is a nice circuit.

Devinder Singh

EFYTimes.com

I have read the 'Shelf Life Of A Techie Is 15 Years!' feature on EFYTimes. It was very informative but I feel that 15 years is applicable to pure IT or software engineering. There is vast difference between a talented experienced engineer and an inexperienced engineer in other branches. Even in electronics/IT hardware, the same may apply. Working principles of capacitors, resistors and inductors are still the same as in 1980s, though sizes may differ. Whereas in pure software or IT, there are many new software and applications being developed every day and so 15 years may hold true. But for a techie from pure electronics and electrical field, this may vary or may not be applicable!

Rakesh
Through e-mail

Circuit Design Software

I visit your EFY website very often from China. I like the circuit section most and would like to know what software or design tools do you use to draw such beautiful circuits diagrams.

Wang Xin
Beijing, China

EFY: Currently we are using gEDA software for making circuits and PCBs. gEDA is an open source software for Linux operating system.

EFY Projects List

I am a final year engineering student and interested in projects related to microcontrollers and embedded systems. Please send me the list of latest projects along with the price list.

Avani Patel
Through e-mail

'Spot An Error' Award Winners

In the 'First look' section of July issue, the specification of MediaPad 10 Link is incorrectly mentioned as 16GB ROM instead of 16GB storage memory.

Niyaz

□ In 'Dual Motor Control for Robots' circuit idea published in July issue, transistors T7, T8, T11 and T12 are indicated as BC338 instead of BC337.

Siddharth Kaul

EFY: Please visit the www.kitsnspares.com website to get the latest projects and prices.

PCB Etching

I tried to make 'Propeller Message Display With Temperature Indicator' circuit published in July issue. But when I took a print of the PCB on a tracing paper and processed it on copper-clad laminate, the copper tracks got separated from the board during etching. Please guide me how I can make PCB for this project.

Kazi Yaser Arafat
Through e-mail

EFY: Looks like the PCB was left in etching solution for too long. Please take precautions mentioned below for a good result:

1. Use a good-quality printer to print the layout on tracing paper
2. Once the layout is printed on copper clad, look for weak spots and darken them using a permanent marker
3. Make the solution with warm water (45 to 50 degrees centigrade)
4. Keep moving the PCB in the solution for quick etching
5. Keep checking whether the etching is complete